ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

SUPPLEMENTARY EXAMINATION — DECEMBER / JUNE 2023

FIRST YEAR (COMMON TO ALL BRANCHES) — SEMESTER 2

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Explain Hardness of water (Temporary vs Permanent). Describe EDTA titration method for hardness estimation. [14 Marks]

[OR]

Q.1 (Alt) Describe Zeolite process and Ion-Exchange demineralization process for water softening. [14 Marks]

--- UNIT II ---

Q.2 Explain Calorific Value of fuels (HCV and LCV). Describe Bomb Calorimeter experiment. [14 Marks]

[OR]

Q.2 (Alt) Explain Proximate and Ultimate analysis of coal. Describe fractional distillation of petroleum. [14 Marks]

--- UNIT III ---

Q.3 Explain Electrochemical theory of Corrosion (Galvanic and Differential Aeration corrosion). [14 Marks]

[OR]

Q.3 (Alt) Describe Corrosion prevention methods: Sacrificial Anode, Impressed Current Cathodic protection, and Metallic coatings. [14 Marks]

--- UNIT IV ---

Q.4 Classify Polymers: Thermoplastic vs Thermosetting. Explain synthesis of Nylon 6,6 and Bakelite. [14 Marks]

[OR]

Q.4 (Alt) Describe Lubricants: Mechanism of lubrication (Fluid film, Boundary, Extreme pressure), Viscosity Index. [14 Marks]

--- UNIT V ---

Q.5 State Beer-Lambert's Law. Explain instrumentation and applications of UV-Visible Spectroscopy. [14 Marks]

[OR]

Q.5 (Alt) Explain Principle and applications of IR Spectroscopy and Phase Rule for 1-component water system. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****